

Filing Lens

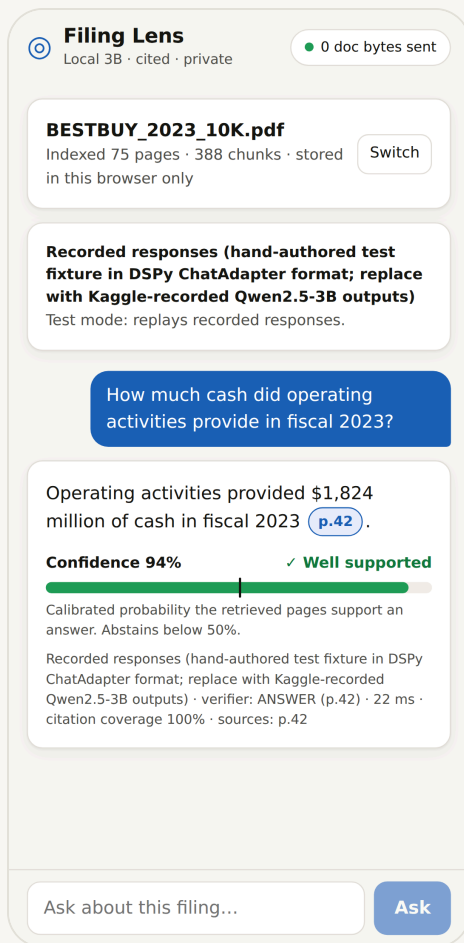
Cited answers from company filings, from a local 3B model (Qwen 2.5, runs on the phone)

Retail investors are handed 100-page 10-Ks. Chatbots answer anyway, with no page reference, and the filing gets uploaded to someone else's server.

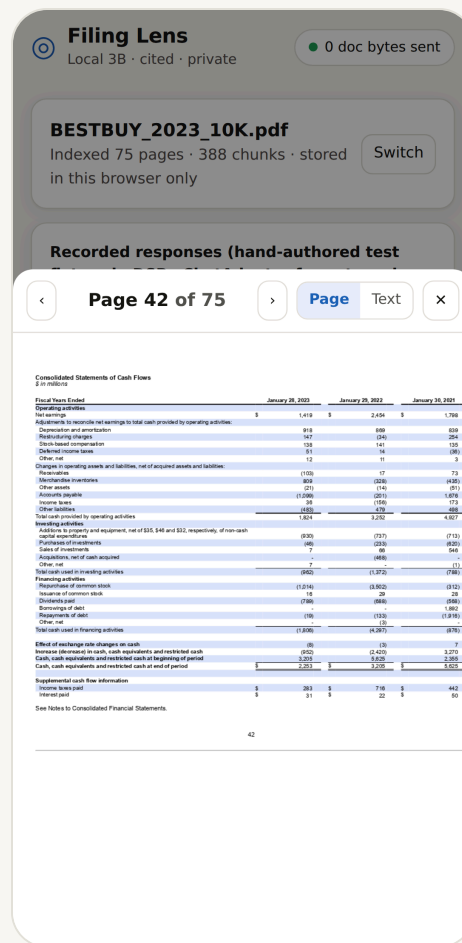
FinanceBench (Islam et al., 2023, arXiv:2311.11944): GPT-4-Turbo with a retrieval system **incorrectly answered or refused 81%** of 150 open-book filing questions.

Problem: wrong answers look the same as right ones, and privacy is the price of asking.

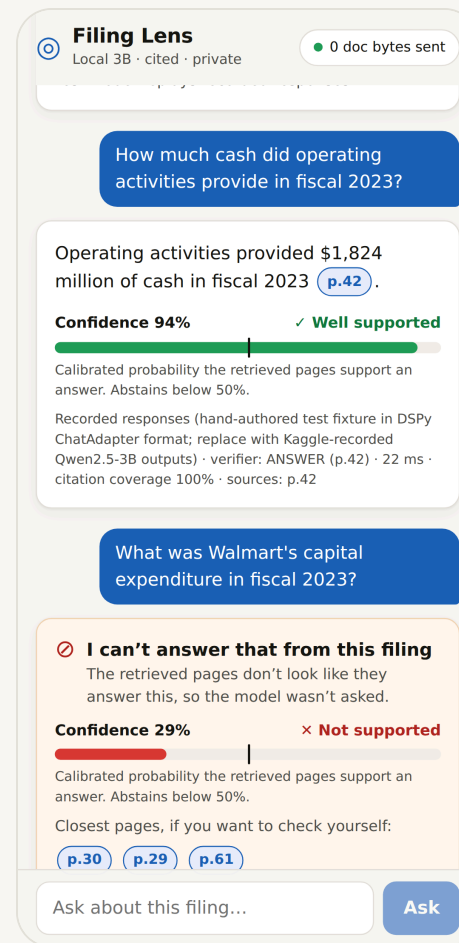
Demo



Cited answer + calibrated confidence



Citation chip opens the page



Wrong-company question → abstain

Recorded in headless Chromium at Pixel-7 size with recorded model responses (the build machine can't download weights). Phone video: [\[ASK: phone recording link\]](#) .

Privacy proof

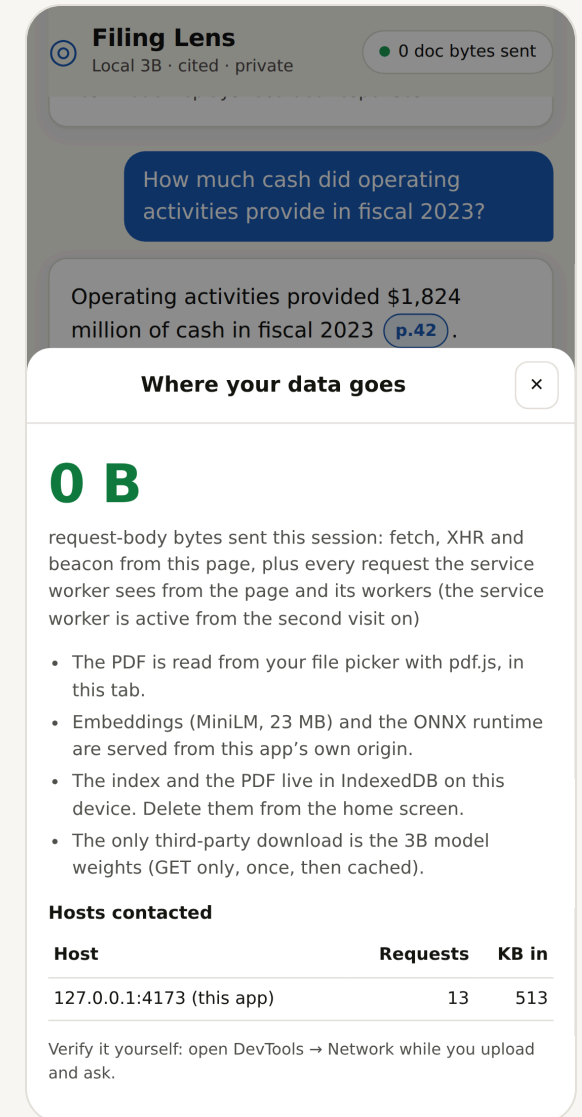
The document never leaves the device. The build's e2e test enforces this; it isn't just claimed.

Requests during upload → index → ask → abstain **20**

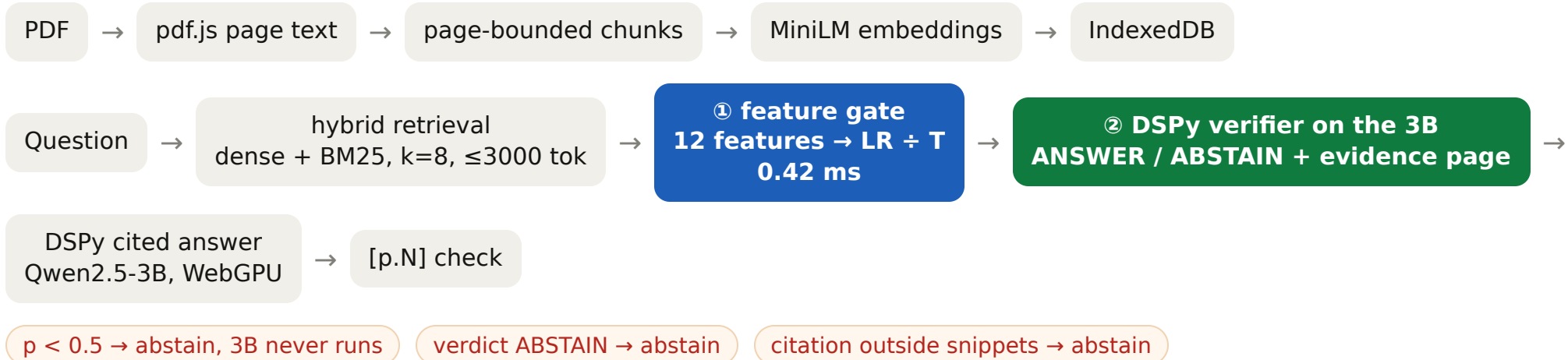
to any host other than the app's own origin **0**

carrying a request body (POST/PUT/beacon) **0**

- pdf.js, MiniLM embeddings, ONNX runtime, wllama: npm packages served from the app origin, no external scripts
- Index + PDF stored in IndexedDB
- Only third-party fetch: the Qwen2.5 weights (GET, once, then cached). Airplane mode after that: **[ASK: confirm on phone]**



Architecture: a two-stage cascade



① is cheap and catches wrong-company questions (it scores retrieval and word coverage, not content). ② reads the actual sentences: it is a DSPy program, compiled with BootstrapFewShot(2 demos); MIPROv2 light tried, dev 0.519 < 0.548, not kept on Qwen/Qwen2.5-3B-Instruct fp16 via llama.cpp (Kaggle T4; vLLM failed to start), exported to `app/prompts/verifier.json` and rebuilt in the browser byte-for-byte. A unit test checks this against DSPy's own rendering.

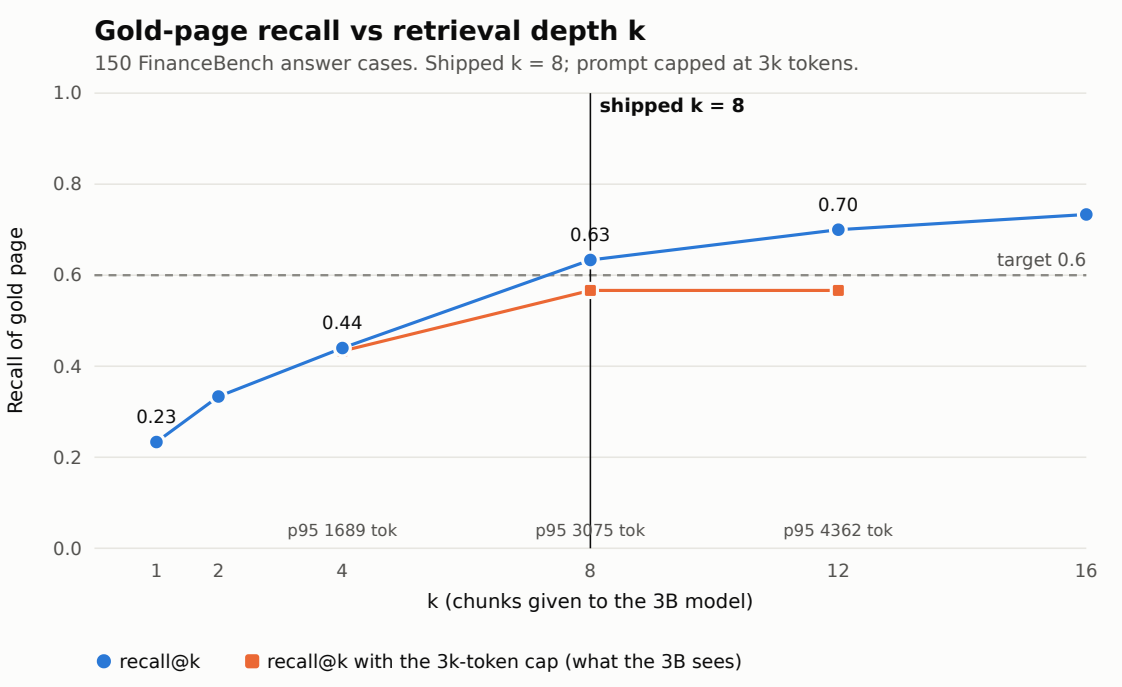
Bench: gate vs verifier vs cascade

FinanceBench open-source (patronus-ai/financebench), 150 questions, 84 filings. Gold answer = question on its own filing; **Easy** abstain = another company's 10-K; **Hard** abstain = own filing with the gold pages (and any page still holding the gold numbers) removed; 10 questions excluded where the answer was still retrieved. Test split (company-grouped), 160 instances.

System	Easy acc.	Hard acc.	ECE E / H	Cite page	p50	3B calls
Feature gate only	0.864	0.524	0.192 / 0.156	0.279*	0.42 ms	0%
Verifier only (fp16_llamacpp)	0.509	0.495	0.476 / 0.496	0.500	7447 ms	100%
Cascade: gate → verifier (fp16_llamacpp)	0.518	0.495	0.392 / 0.404	0.500	7202 ms	53%
Verifier only (gguf_q4)	0.527	0.495	0.468 / 0.474	0.500	7156 ms	100%
Cascade: gate → verifier (gguf_q4)	0.536	0.495	0.410 / 0.391	0.500	6939 ms	53%

* gate proposes no page: top-ranked retrieved page. Gate p50 measured in Node on Node v22.22.0, 4 vCPU container, no GPU; verifier p50: one verifier call on a Kaggle T4 (llama.cpp; vLLM failed to start).

Gate, all 440 out-of-fold: Easy acc 0.870, abstain P/R 0.828 / 0.933; Hard acc 0.524. Two "does the top passage answer this?" features: Hard 0.528 → 0.524.



Limits (measured)

- **The feature gate can't see content: Hard 0.524** (always answering scores 0.517 on this set). It scores retrieval and word overlap, so "the gold page" and "a nearby page about the same metric" look alike. The verifier stage was meant for that case. It didn't help: the compiled 3B verifier abstains on almost everything (Hard 0.495, Easy 0.527 on Q4; auto-graded answer accuracy 0.036). Behind the gate it cuts Easy accuracy from 0.864 to 0.536, so it ships off by default, as an opt-in Strict mode.
- **Retrieval is the bottleneck:** gold-page recall@4 0.440, @8 0.633, @12 0.700. With the 3k-token cap the 3B sees the gold page 57% of the time; the sentence snippets keep it 53%.
- **A failure I shipped and fixed:** Best Buy's 10-K never mentions Walmart, yet "Walmart capex?" passed the first gate. Adding a feature for whether the companies a question names appear in the filing at all fixed it (gate accuracy 0.82 → 0.91 on the earlier own-vs-wrong-company bench).
- Answers are auto-graded (number within 1%, else token-F1), an approximation of FinanceBench's human grading. Scanned PDFs (no text layer) are rejected.

Roadmap

- Verifier: 2 bootstrapped demos on a 3B model learned to abstain. Next: more labelled demos, a threshold on $P(\text{ANSWER})$ tuned on dev rather than the argmax verdict, and a 7B check on a laptop GPU
- Retrieval: table-aware chunking for financial statements; a reranker only if one ships as an npm package (none found) and adds <150 ms
- Raise the context budget if the phone allows (uncapped k=8 prompts reach 3586 tokens)
- Indian filings: BSE/NSE annual reports, Hindi questions

The ask

A Finale slot, one iQOO test phone to benchmark on-device inference, and mentor feedback on the Snapdragon NPU path.

Repo: github.com/rajyyug1132/Filing-Lens-Private-Cited-Answers

Cited answers, or an honest "I can't tell". Your filing stays yours.